

Assignment 3: Data Exploration

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Exploration M3: Data Exploration.

Directions

1. Rename this file `<FirstLast>_A03_DataExploration.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Assign a useful **name to each code chunk** and include ample **comments** with your code.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.
7. After Knitting, submit the completed exercise (PDF file) to the dropbox in Canvas.

TIP: If your code extends past the page when knit, tidy your code by manually inserting line breaks.

TIP: If your code fails to knit, check that no `install.packages()` or `View()` commands exist in your code.

Set up your R session

1. Load necessary packages (tidyverse, lubridate, here), check your current working directory and upload two datasets: the ECOTOX neonicotinoid dataset (ECOTOX_Neonicotinoids_Insects_raw.csv) and the Niwot Ridge NEON dataset for litter and woody debris (NEON_NIWO_Litter_massdata_2016-26_raw.csv). Name these datasets “Neonics” and “Litter”, respectively.

```
#load packages
library(tidyverse)
library(lubridate)
library(here)

#Checking my current working directory
here()
```

```
## [1] "/home/guest/EDE_R_Lab_F26_NEW"
```

```
cat("\\newpage")
```

```
## \newpage
```

```
#Importing my datasets
Neonics <- read.csv(
  file = here("Data/Raw/ECOTOX_Neonicotinoids_Insects_raw.txt"),
  sep = "|"
)

Litter <- read.csv(
  file = here("Data/Raw/NEON_NIWO_Litter_massdata_2016-26_raw.csv")
)
```

Learn about your system

2. The neonicotinoid dataset was collected from the Environmental Protection Agency’s ECOTOX Knowledgebase, a database for ecotoxicology research. Neonicotinoids are a class of insecticides used widely in agriculture. The dataset that has been pulled includes all studies published on insects. Why might we be interested in the ecotoxicology of neonicotinoids on insects? Feel free to do a brief internet search if you feel you need more background information.

Answer: I think we’d be interested in the ecotoxicology of neonicotinoids for two reasons. First, I think we would want to understand their effectiveness in controlling the agriculture pests they are designed to address. Second, I understand that neonicotinoids can harm honeybees and other pollinators, and we would want to better understand that impact on those insects.

3. The Niwot Ridge litter and woody debris dataset was collected from the National Ecological Observatory Network, which collectively includes 81 aquatic and terrestrial sites across 20 ecoclimatic domains. 32 of these sites sample forest litter and woody debris, and we will focus on the Niwot Ridge long-term ecological research (LTER) station in Colorado. Why might we be interested in studying litter and woody debris that falls to the ground in forests? Feel free to do a brief internet search if you feel you need more background information.

Answer: I think this is of interest as wildfires are becoming more intense. Litter and woody debris are one contributing factor, and it would be helpful to study that connection and monitor the amount of debris. Litter and woody debris are also important in an ecosystem’s nutrient cycling and provide important wildlife habitat. Monitoring it could help us understand the health of that ecosystem.

4. How is litter and woody debris sampled as part of the NEON network? Read the [NEON_Litterfall_UserGuide.pdf](#) document to learn more. List three pieces of salient information about the sampling methods here:

Answer: 1. Sampling occurs only in Tower plots, and their locations were randomly selected “within the 90% flux footprint of the primary and secondary airsheds.” 2. Sampling at forested tower airshed sites occurs “in up to two randomly selected subplots in each of 20 40m x 40m plots” while low-saturated vegetation sites have “up to 30 20m x 20m plots.” The number of plots can sometimes be lower if space doesn’t allow. 3. For every 400 square meters plot area, there is one elevated trap and/or one ground trap. Ground traps are sampled once per year. Elevated trap sampling is dependent upon vegetation with sampling occurring once every 2 weeks in deciduous forests and once every 1 to 2 months in coniferous forests.

Obtain basic summaries of your data (Neonics)

5. What are the dimensions of the dataset?

```
#Finding the dimensions of the Neonics dataset with the dim function
dim(Neonics)
```

```
## [1] 8748 76
```

6. Using the `summary` function on the “Effect” column, determine the most common effects that are studied. Why might these effects specifically be of interest? [Tip: The `sort()` command is useful for listing the values in order of magnitude...]

```
#Using the summary function on the Effect column to determine the most common effects
summary(Neonics$Effect)
```

```
##      Length      Class      Mode
##      8748 character character
```

```
#Summary function only told me the length, class, and mode information rather than the actual effects.
Neonics$Effect <- factor(Neonics$Effect)
summary(Neonics$Effect)
```

```
##      Accumulation      Avoidance      Behavior      Biochemistry
##           47           86           826           151
##      Cell(s)      Development      Enzyme(s) Feeding behavior
##           57          499           405           712
##      Genetics      Growth      Histology      Hormone(s)
##          905          153           20           8
##      Immunological      Injury      Intoxication      Morphology
##           38           2           52           87
##      Mortality      Physiology      Population      Reproduction
##          2687          79          1377          557
```

Answer: The four most common effects are Mortality, Population, Genetics, and Behavior. These are particularly interesting because they are high impact effects. The insect dying and changes in population, genetics, and behavior all have significant meaning for that species and potentially, the ecosystem.

7. Using the `summary` function, determine the six most commonly studied species in the dataset (common name). What do these species have in common, and why might they be of interest over other insects? Feel free to do a brief internet search for more information if needed.[TIP: Explore the help on the `summary()` function, in particular the `maxsum` argument...]

```
#Using the factor function first to convert the text into data that is sortable and then using the sum
Neonics$X.Species.Common.Name <-factor(Neonics$X.Species.Common.Name)
summary(Neonics$X.Species.Common.Name)
```

```
##      Honey Bee      Carniolan Honey Bee
##           2310           622
##      Parasitic Wasp      Bumble Bee
##           428           360
##      Italian Honeybee      Asiatic Honey Bee
##           348           345
```

##	Sevenspotted Lady Beetle	Buff Tailed Bumblebee
##	270	216
##	Monarch Butterfly	Buff-tailed Bumblebee
##	194	169
##	Silkworm	Mexican Beetle
##	167	153
##	Bee	Predatory Mite
##	114	114
##	Ladybird Beetle	Parasitoid Wasp
##	113	109
##	Stingless Bee	House Fly
##	92	89
##	Sweetpotato Whitefly	Common Green Lacewing
##	83	76
##	Red Mason Bee	Aphid Family
##	74	71
##	Chalcid Wasp	Predatory Bug
##	71	70
##	Macedonian Honey Bee	Minute Pirate Bug
##	65	63
##	Braconid Wasp	Araneoid Spider Order
##	61	54
##	Wasp	Thrip Order
##	53	50
##	Cotton Aphid	Turnip Aphid
##	41	41
##	Wireworm	Wolf Spider
##	41	41
##	Eulophid Parasitoid	Aphid Wasp
##	40	34
##	Asian Lady Beetle	Azalea Lace Bug
##	34	34
##	Corn Leaf Aphid	Alfalfa Leafcutter Bee
##	32	30
##	Cabbage Aphid	Endoparasitoid
##	30	30
##	Mirid Bug	Thrips
##	30	30
##	Convergent Lady Beetle	Mite
##	29	29
##	Pistachio Psyllid	Sand Wasp
##	29	29
##	Green Lacewing	Asian Citrus Psyllid
##	28	27
##	Cabbage Stem Flea Beetle	Solenopsis Mealybug
##	27	26
##	Moth	Beetle Order
##	25	24
##	Ladybird Beetle Family	Citrus Leafminer
##	23	22
##	Ground Dwelling Spider	Lacewing
##	22	22
##	Weaver Spider	Aphid
##	22	18

##	Linyphiid Spider	Samurai wasp
##	18	18
##	Soldier Beetle	Spotless Ladybird Beetle
##	18	18
##	Mason Bee	Colorado Potato Beetle
##	17	16
##	English Grain Aphid	Western Flower Thrips
##	16	16
##	Black-spotted Lady Beetle	Dwarf Honey Bee
##	15	15
##	Eastern Subterranean Termite	Two-Spotted Spider Mite
##	15	15
##	Wireworms	Bagrada Bug
##	15	14
##	Blue Gum Chalcid Wasp	Budworm
##	14	14
##	Parasitoid	Chilean Predatory Mite
##	14	13
##	Giant Honeybee	Neotropical Stingless Bee
##	13	13
##	Painted Lady	Thrip
##	13	13
##	Brown Planthopper	Chinch Bug
##	12	12
##	Leafhopper	Mosquito
##	12	12
##	White-backed Planthopper	Tobacco Budworm Parasitoid Wasp
##	12	11
##	Click Beetle Family	Codling Moth
##	10	10
##	Glasshouse Potato Wasp	Hover Fly
##	10	10
##	Jassid	Mulberry Thrip
##	10	10
##	Southern House Mosquito	Transverse Ladybird Beetle
##	10	10
##	Turf Running-spider	Gall Wasps
##	10	9
##	Minute Parasitic Wasps	(Other)
##	9	342

```
summary(Neonics$X.Species.Common.Name, maxsum = 6)
```

##	Honey Bee Carniolan Honey Bee	Parasitic Wasp	Bumble Bee
##	2310	622	428
##	Italian Honeybee	(Other)	360
##	348	4680	

```
summary(Neonics$X.Species.Common.Name, maxsum = 7)
```

##	Honey Bee Carniolan Honey Bee	Parasitic Wasp	Bumble Bee
##	2310	622	428
##	Italian Honeybee	Asiatic Honey Bee	(Other)
##	348	345	4335

#After using the summary feature initially, I tried out maxsum. When I selected 6, it grouped all of the

Answer: The six most commonly studied species are honey bee, Carniolan honey bee, parasitic wasp, bumble bee, Italian honeybee, and Asiatic honey bee. All of these species are pollinators, which would make them of great interest to study.

8. Concentrations are always a numeric value. What is the class of `Conc.1.Mean..Author..` column in the dataset, and why is it not numeric? [Tip: Viewing the dataframe may be helpful...]

```
#Viewing the dataframe to better understand the data in Conc.1.Mean..Author.. and then checking the class  
#view(Neonics)  
class(Neonics$X.Conc.1.Mean..Author..)
```

```
## [1] "character"
```

#commenting out view to avoid knitting issues

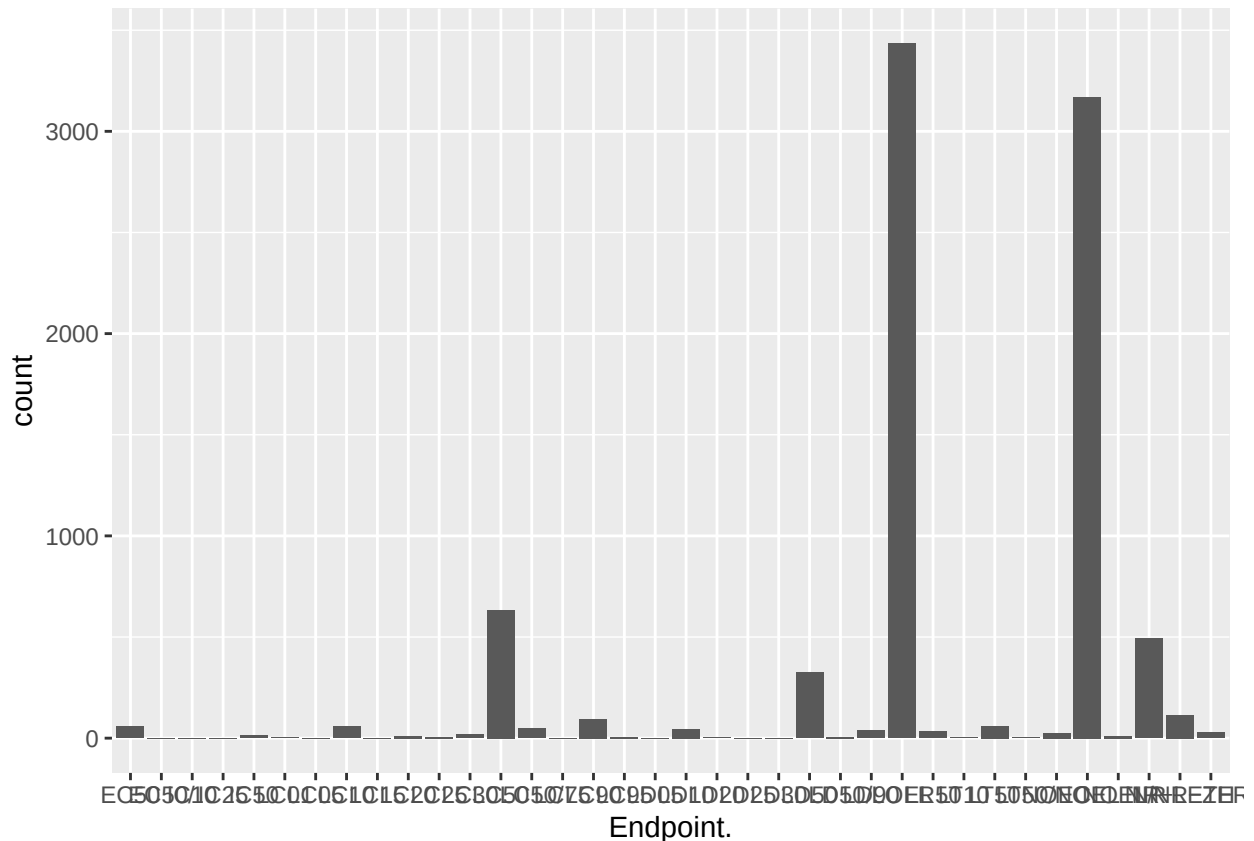
Answer: The class of `Conc.1.Mean..Author..` is character. I believe this may be the case because some of the numbers have a / following the number.

Explore your data graphically (Neonics)

9. Create a bar graph of Endpoint counts. What are the two most common end points, and how are they defined? Consult the ECOTOX_CodeAppendix for more information.

[**TIP:** Add `theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))` to the end of your plot command to rotate and align the X-axis labels...]

```
ggplot(Neonics, aes(x = Endpoint.)), theme(axis.tent.x = element_text(angle = 90, vjust = 0.5, hjust = 1.  
geom_bar()
```



#I created a bar graph using the endpoint column data, including axis labels.

Answer: The two most common endpoints are LOEL and NOEL. LOEL represents the lowest-observable-effect-level, which means the lowest dose or concentration that produces effects that are significantly different than the control group. NOEL represents no-observable-effect level, which means the highest dose or concentration that produces effects that are not significantly different than the control group.

Explore your data (Litter)

10. Determine the class of `collectDate`. Is it a date? If not, change to a date and confirm the new class of the variable. Using the `unique` function, determine which dates litter was sampled in August 2018.

```
#view(Litter)
#I viewed the Litter data and then commented out the function to avoid knitting issues.
#Using class on collectDate as a starting point
class(Litter$collectDate)
```

```
## [1] "character"
```

```
cat("\\newpage")
```

\newpage

```
#Since collectDate is considered a character rather than a date, I'm using lubridate to change it to date
Litter$collectDate <- ymd(Litter$collectDate)
#Confirming with class that it is now in date format
class(Litter$collectDate)
```

```
## [1] "Date"
```

```
#Using unique to determine which dates litter was sampled in August 2018.
unique(Litter$collectDate)
```

```
## [1] "2016-06-16" "2016-07-14" "2016-08-11" "2016-09-08" "2016-10-07"
## [6] "2016-11-03" "2017-07-06" "2017-07-07" "2017-08-03" "2017-08-31"
## [11] "2017-10-12" "2017-11-20" "2018-06-07" "2018-07-05" "2018-08-02"
## [16] "2018-08-30" "2018-09-27" "2018-10-25" "2018-11-21" "2019-06-05"
## [21] "2019-07-02" "2019-07-31" "2019-08-28" "2019-09-25" "2019-10-23"
## [26] "2020-07-01" "2020-07-29" "2020-08-26" "2020-09-22" "2020-11-18"
## [31] "2021-06-28" "2021-07-27" "2021-08-23" "2021-09-20" "2021-09-22"
## [36] "2021-10-18" "2022-06-16" "2022-06-30" "2022-07-27" "2022-08-23"
## [41] "2022-09-19" "2022-10-17" "2023-06-05" "2023-06-28" "2023-07-24"
## [46] "2023-08-24" "2023-09-19" "2023-10-17" "2023-11-13" "2023-12-11"
## [51] "2024-06-25" "2024-07-23" "2024-08-19" "2024-09-16" "2024-09-18"
## [56] "2024-10-14"
```

```
#Litter was sampled on August 2, 2018, and August 30, 2018.
```

11. Using the `unique` function, list the different plotIDs sampled at Niwot Ridge. How is the information obtained from `unique` different from that obtained from `summary`?

```
#Using the unique function to obtain the list of different plotIDs sampled at Niwot Ridge.
unique(Litter$plotID)
```

```
## [1] "NIWO_047" "NIWO_061" "NIWO_062" "NIWO_057" "NIWO_058" "NIWO_064"
## [7] "NIWO_051" "NIWO_067" "NIWO_063" "NIWO_041" "NIWO_046" "NIWO_040"
```

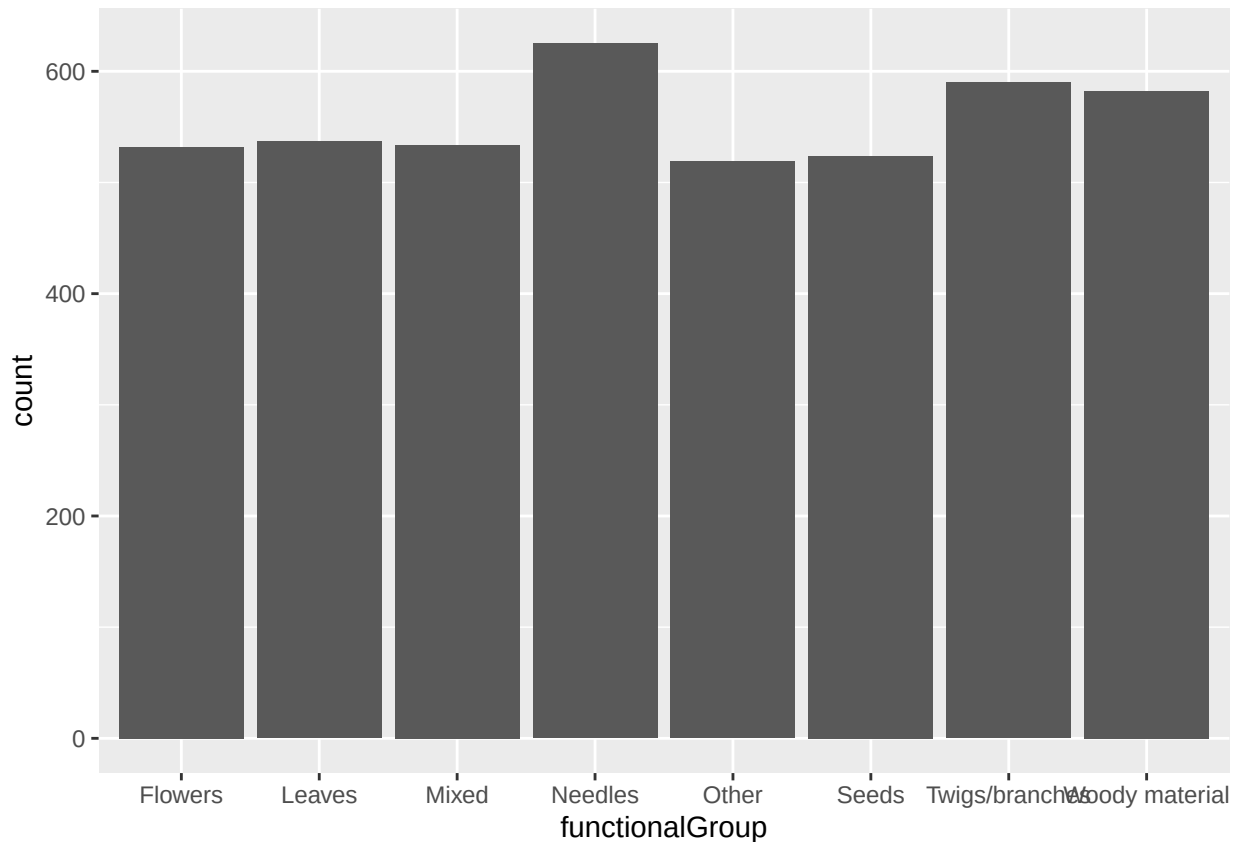
```
#Using the summary function as a comparison.
summary(Litter$plotID)
```

```
##      Length      Class      Mode
##      4443 character character
```

Answer: The `unique` function returned the various plotIDs sampled outright. The `summary` function provided the length (4443), class (character) and mode (character). In order to obtain the same information that the `unique` function provide, I would have needed to factor the plotID data first and then use `summary`.

12. Create a bar graph of functionalGroup counts. This shows you what type of litter is collected at the Niwot Ridge sites. Notice that litter types are fairly equally distributed across the Niwot Ridge sites.

```
#Creating a bar graph to visually review the functionalGroup data  
ggplot(Litter, aes(x = functionalGroup)) +  
  geom_bar()
```



What type(s) of litter tend to have the highest biomass at these sites?

Answer: Based on the functionalGroup bar graph, the litter with the highest biomass seems to be from trees. Needles was the number one group, followed by twigs/branches, and then woody material.

13. Write a short data-exploration summary for Neonics.

- **One key characteristic:** Data from the U.S. EPA collected between 2015-2026 with 8,748 entries about the response of insects to neonicotinoids
- **One data-quality issue:** The endpoint bar graph demonstrated that there may be some coding issues with the data. For instance, EC50 and EC50/ are separate bars, and the addition of slashes occurs with several endpoint categories.
- **One interesting pattern:** Six pollinators are the most commonly studied species
- **One question for further analysis:** Is there a relationship between the most commonly studied species and the most common effects?

14. Write a short data-exploration summary for Litter.

- **One key characteristic:** Data from NEON of litter and woody debris samples collected between January 2016-December 2024 at the Niwot Ridge long-term ecological research station
- **One data-quality issue:** I noticed when viewing the data in the remarks field, there are quite a few instances where there was no sample to collect or not enough data to register. It may be helpful to look for trends or see what percentage of the time this was the case.
- **One interesting pattern:** The litter types are fairly equally distributed.
- **One question for further analysis:** Are there seasonal or annual trends with litter type?